

CS4221/5421: Tutorial 7 — Functional Dependencies

Pratik Karmakar

School of Computing,
National University of Singapore

AY25/26 S1



Identifying Functional Dependencies

Example Relation, $R = \{A, B, C\}$ and $\Sigma = \{\{A\} \rightarrow \{B\}, \{B\} \rightarrow \{C\}\}$

The column in **red** decides the color of the row: The rows having same values in the **red** column, have same color.

Task: Check which other columns are following "same value if same color" for each deciding (**red**) column. The columns for which this holds, are implied by the **red** column.

A	B	C
a1	b1	c1
a1	b1	c1
a2	b2	c2
a2	b2	c2
a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
a5	b2	c2

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$\{A\} \rightarrow \{A, B, C\}$

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$\{A\} \rightarrow \{A, B, C\}$

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a2	b2	c2
a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
a5	b2	c2

$\{B\} \rightarrow \{B, C\}$

Identifying Functional Dependencies

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a4	b1	c1
a5	b2	c2

$\{A\} \rightarrow \{A, B, C\}$

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a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
a5	b2	c2

$\{B\} \rightarrow \{B, C\}$

A	B	C
a1	b1	c1
a1	b1	c1
a2	b2	c2
a2	b2	c2
a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
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a5	b2	c2

$\{A\} \rightarrow \{A, B, C\}$

A	B	C
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a1	b1	c1
a2	b2	c2
a2	b2	c2
a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
a5	b2	c2

$\{B\} \rightarrow \{B, C\}$

A	B	C
a1	b1	c1
a1	b1	c1
a2	b2	c2
a2	b2	c2
a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
a5	b2	c2

$\{C\} \rightarrow \{C\}$

Rules for Functional Dependencies (Armstrong's Axioms)

W, X, Y, Z mentioned here are **sets** of attributes.

Sound & complete inference system

- **Reflexivity:** If $Y \subseteq X$, then $X \rightarrow Y$.
- **Augmentation:** If $X \rightarrow Y$, then $X \cup Z \rightarrow Y \cup Z$ for any Z .
- **Transitivity:** If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$.

Common derived rules (from the three above)

- **Union / Additivity:** If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow Y \cup Z$.
- **Decomposition / Projectivity:** If $X \rightarrow Y \cup Z$, then $X \rightarrow Y$ and $X \rightarrow Z$.
- **Pseudotransitivity:** If $X \rightarrow Y$ and $W \cup Y \rightarrow Z$, then $W \cup X \rightarrow Z$.
- **Composition:** If $X \rightarrow Y$ and $Z \rightarrow W$, then $X \cup Z \rightarrow Y \cup W$.
- **Self-determination:** $X \rightarrow X$.

How we use them here

- *LHS reduction:* test if $X \setminus \{A\} \rightarrow Y$ holds via closure using the axioms.
- *Redundancy removal:* test if an FD $X \rightarrow Y$ is implied by the rest (i.e., $Y \subseteq X^+$ computed from the others).

Question 1.a.

Consider the following schema R and the set of functional dependencies Σ .

$$R = \{A, B, C, D, E\},$$

$$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}.$$

Question 1.a.

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Show that $\Sigma \vdash \{C, D\} \rightarrow \{A, B, C, D\}$ using Armstrong's axioms.

- 1 $\{C, D\} \rightarrow \{A, B\}$
- 2 $\{A, B, C\} \rightarrow \{E\}$
- 3 $\{B, C, D\} \rightarrow \{A, E\}$
- 4 $\{B, D\} \rightarrow \{A\}$
- 5 $\{A, C\} \rightarrow \{B, E\}$

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Show that $\Sigma \vdash \{C, D\} \rightarrow \{A, B, C, D\}$ using Armstrong's axioms.

$$\textcircled{6} \{C, D\} \rightarrow \{A, B, C, D\} \text{ (Augmentation of (1) with } \{C, D\})$$

$$\textcircled{1} \{C, D\} \rightarrow \{A, B\}$$

$$\textcircled{2} \{A, B, C\} \rightarrow \{E\}$$

$$\textcircled{3} \{B, C, D\} \rightarrow \{A, E\}$$

$$\textcircled{4} \{B, D\} \rightarrow \{A\}$$

$$\textcircled{5} \{A, C\} \rightarrow \{B, E\}$$

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$$R = \{A, B, C, D, E\},$$

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- ① $\{C, D\} \rightarrow \{A, B\}$
- ② $\{A, B, C\} \rightarrow \{E\}$
- ③ $\{B, C, D\} \rightarrow \{A, E\}$
- ④ $\{B, D\} \rightarrow \{A\}$
- ⑤ $\{A, C\} \rightarrow \{B, E\}$
- ⑥ $\{C, D\} \rightarrow \{A, B, C, D\}$ (Augmentation of (1) with $\{C, D\}$)
- ⑦ $\{B, C, D\} \rightarrow \{A, B, C, D, E\}$ (Augmentation of (3) with $\{B, C, D\}$)

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- ⑦ $\{B, C, D\} \rightarrow \{A, B, C, D, E\}$ (Augmentation of (3) with $\{B, C, D\}$)
- ⑧ $\{A, B, C, D\} \rightarrow \{B, C, D\}$ (Reflexivity)

Question 1.a.

Consider the following schema R and the set of functional dependencies Σ .

$$R = \{A, B, C, D, E\},$$

$$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}.$$

Show that $\Sigma \vdash \{C, D\} \rightarrow \{A, B, C, D\}$ using Armstrong's axioms.

- ① $\{C, D\} \rightarrow \{A, B\}$
- ② $\{A, B, C\} \rightarrow \{E\}$
- ③ $\{B, C, D\} \rightarrow \{A, E\}$
- ④ $\{B, D\} \rightarrow \{A\}$
- ⑤ $\{A, C\} \rightarrow \{B, E\}$
- ⑥ $\{C, D\} \rightarrow \{A, B, C, D\}$ (Augmentation of (1) with $\{C, D\}$)
- ⑦ $\{B, C, D\} \rightarrow \{A, B, C, D, E\}$ (Augmentation of (3) with $\{B, C, D\}$)
- ⑧ $\{A, B, C, D\} \rightarrow \{B, C, D\}$ (Reflexivity)
- ⑨ $\{C, D\} \rightarrow \{B, C, D\}$ (Transitivity between (6) and (8))

Question 1.a.

Consider the following schema R and the set of functional dependencies Σ .

$$R = \{A, B, C, D, E\},$$

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- ① $\{C, D\} \rightarrow \{A, B\}$
- ② $\{A, B, C\} \rightarrow \{E\}$
- ③ $\{B, C, D\} \rightarrow \{A, E\}$
- ④ $\{B, D\} \rightarrow \{A\}$
- ⑤ $\{A, C\} \rightarrow \{B, E\}$
- ⑥ $\{C, D\} \rightarrow \{A, B, C, D\}$ (Augmentation of (1) with $\{C, D\}$)
- ⑦ $\{B, C, D\} \rightarrow \{A, B, C, D, E\}$ (Augmentation of (3) with $\{B, C, D\}$)
- ⑧ $\{A, B, C, D\} \rightarrow \{B, C, D\}$ (Reflexivity)
- ⑨ $\{C, D\} \rightarrow \{B, C, D\}$ (Transitivity between (6) and (8))
- ⑩ $\{C, D\} \rightarrow \{A, B, C, D, E\}$ (Transitivity between (9) and (7))

Question 1.b.

$$R = \{A, B, C, D, E\},$$

$$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}.$$

Show that $\Sigma \models \{C, D\} \rightarrow \{A, B, E\}$ using **The Chase**.

Question 1.b.

$$R = \{A, B, C, D, E\},$$

$$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}.$$

Show that $\Sigma \models \{C, D\} \rightarrow \{A, B, E\}$ using **The Chase**.

A	B	C	D	E
α	α	α	α	α
a_1	b_1	c_1	d_1	e_1

Setup

Question 1.b.

$$R = \{A, B, C, D, E\},$$

$$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}.$$

Show that $\Sigma \models \{C, D\} \rightarrow \{A, B, E\}$ using **The Chase**.

A	B	C	D	E
α	α	α	α	α
a_1	b_1	c_1	d_1	e_1

Setup

A	B	C	D	E
α	α	α	α	α
a_1	b_1	α	α	e_1

Initialization

Question 1.b.

$$R = \{A, B, C, D, E\},$$

$$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}.$$

Show that $\Sigma \models \{C, D\} \rightarrow \{A, B, E\}$ using **The Chase**.

A	B	C	D	E
α	α	α	α	α
a_1	b_1	c_1	d_1	e_1

Setup

A	B	C	D	E
α	α	α	α	α
a_1	b_1	α	α	e_1

Initialization

A	B	C	D	E
α	α	α	α	α
α	α	α	α	e_1

 $\{C, D\} \rightarrow \{A, B\}$

Question 1.b.

$$R = \{A, B, C, D, E\},$$

$$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}.$$

Show that $\Sigma \models \{C, D\} \rightarrow \{A, B, E\}$ using **The Chase**.

A	B	C	D	E
α	α	α	α	α
a_1	b_1	c_1	d_1	e_1

Setup

A	B	C	D	E
α	α	α	α	α
a_1	b_1	α	α	e_1

Initialization

A	B	C	D	E
α	α	α	α	α
α	α	α	α	e_1

 $\{C, D\} \rightarrow \{A, B\}$

A	B	C	D	E
α	α	α	α	α
α	α	α	α	α

 $\{A, C, D\} \rightarrow \{E\}$

We stop here as we have already reached $\{C, D\} \rightarrow \{A, B, E\}$ (the values in the second row to look at).

Question 1.c.

Show that $\{A, C\} \rightarrow \{D\}$ does not hold on R with Σ using Algorithm 1.

Algorithm 1: Attribute Closure Algorithm

Input: S, Σ

Output: S^+

$\Omega \leftarrow \Sigma;$

// unused

$\Gamma \leftarrow S;$

// closure

while *there exists* $(X \rightarrow Y) \in \Omega$ *such that* $X \subseteq \Gamma$ **do**

$\Omega \leftarrow \Omega \setminus \{X \rightarrow Y\};$

$\Gamma \leftarrow \Gamma \cup Y;$

return $\Gamma;$

Using Algorithm 1 we get $\{A, C\}^+ = \{A, B, C, E\}$. As $D \notin \{A, C\}^+$, $\{A, C\} \nrightarrow \{D\}$

From Σ we get $\{A, C\} \rightarrow \{B, E\}$. $\{B\}$ determines nothing other than $\{B\}$. Same holds for $\{E\}$. Also, we see $\{B, E\}$ determines nothing other than $\{B, E\}$.

Question 1.d.

Find all keys of R w.r.t. Σ .

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Find all keys of R w.r.t. Σ . If we go algorithmically, we need to find the closures of all subsets of R .

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$$\{A\}^+ = \{A\}$$

$$\{B, E\}^+ = \{B, E\}$$

$$\{B, C, E\}^+ = \{B, C, E\}$$

$$\{B\}^+ = \{B\}$$

$$\{C, D\}^+ = \{A, B, C, D, E\}$$

$$\{B, D, E\}^+ = \{A, B, D, E\}$$

$$\{C\}^+ = \{C\}$$

$$\{C, E\}^+ = \{C, E\}$$

$$\{C, D, E\}^+ = \{A, B, C, D, E\}$$

$$\{D\}^+ = \{D\}$$

$$\{D, E\}^+ = \{D, E\}$$

$$\{A, B, C, D\}^+ = \{A, B, C, D, E\}$$

$$\{E\}^+ = \{E\}$$

$$\{A, B, C\}^+ = \{A, B, C, E\}$$

$$\{A, B, C, E\}^+ = \{A, B, C, E\}$$

$$\{A, B\}^+ = \{A, B\}$$

$$\{A, B, D\}^+ = \{A, B, D\}$$

$$\{A, B, D, E\}^+ = \{A, B, D, E\}$$

$$\{A, C\}^+ = \{A, B, C, E\}$$

$$\{A, B, E\}^+ = \{A, B, E\}$$

$$\{A, D\}^+ = \{A, D\}$$

$$\{A, C, D\}^+ = \{A, B, C, D, E\}$$

$$\{A, C, D, E\}^+ = \{A, B, C, D, E\}$$

$$\{A, E\}^+ = \{A, E\}$$

$$\{A, C, E\}^+ = \{A, B, C, E\}$$

$$\{B, C, D, E\}^+ = \{A, B, C, D, E\}$$

$$\{B, C\}^+ = \{B, C\}$$

$$\{A, D, E\}^+ = \{A, D, E\}$$

$$\{B, D\}^+ = \{A, B, D\}$$

$$\{B, C, D\}^+ = \{A, B, C, D, E\}$$

$$\{A, B, C, D, E\}^+ = \{A, B, C, D, E\}$$

Question 1.e.

Find one minimal cover of $\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}$

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Step 1: Simplify the RHS

- ① $\{C, D\} \rightarrow \{A\}$
- ② $\{C, D\} \rightarrow \{B\}$
- ③ $\{A, B, C\} \rightarrow \{E\}$
- ④ $\{B, C, D\} \rightarrow \{A\}$
- ⑤ $\{B, C, D\} \rightarrow \{E\}$
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Step 1: Simplify the RHS

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- ⑧ $\{A, C\} \rightarrow \{E\}$

Step 2: Simplify the LHS

- ⑨ $\{C, D\} \rightarrow \{A\}$
- ⑩ $\{C, D\} \rightarrow \{B\}$
- ⑪ $\{A, C\} \rightarrow \{E\}$ From (8)
- ⑫ $\{B, D\} \rightarrow \{A\}$ From (6)
- ⑬ $\{C, D\} \rightarrow \{E\}$ From $\{C, D\}^+$
- ⑭ $\{B, D\} \rightarrow \{A\}$
- ⑮ $\{A, C\} \rightarrow \{B\}$
- ⑯ $\{A, C\} \rightarrow \{E\}$

Question 1.e.

Find one minimal cover of $\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}$

Step 1: Simplify the RHS

- 1 $\{C, D\} \rightarrow \{A\}$
- 2 $\{C, D\} \rightarrow \{B\}$
- 3 $\{A, B, C\} \rightarrow \{E\}$
- 4 $\{B, C, D\} \rightarrow \{A\}$
- 5 $\{B, C, D\} \rightarrow \{E\}$
- 6 $\{B, D\} \rightarrow \{A\}$
- 7 $\{A, C\} \rightarrow \{B\}$
- 8 $\{A, C\} \rightarrow \{E\}$

Step 2: Simplify the LHS

- 9 $\{C, D\} \rightarrow \{A\}$
- 10 $\{C, D\} \rightarrow \{B\}$
- 11 $\{A, C\} \rightarrow \{E\}$ From (8)
- 12 $\{B, D\} \rightarrow \{A\}$ From (6)
- 13 $\{C, D\} \rightarrow \{E\}$ From $\{C, D\}^+$
- 14 $\{B, D\} \rightarrow \{A\}$
- 15 $\{A, C\} \rightarrow \{B\}$
- 16 $\{A, C\} \rightarrow \{E\}$

Remove redundancies

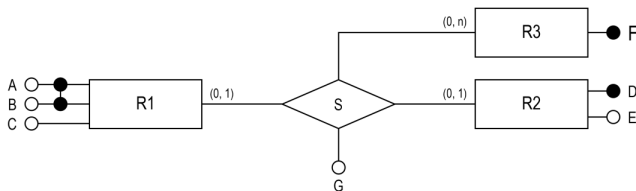
- 17 ~~$\{C, D\} \rightarrow \{A\}$~~ As it can be derived from (10) and (14)
- 18 $\{C, D\} \rightarrow \{B\}$
- 19 ~~$\{A, C\} \rightarrow \{E\}$~~ As we have it in (16)
- 20 ~~$\{B, D\} \rightarrow \{A\}$~~ As we have it in (14)
- 21 ~~$\{C, D\} \rightarrow \{E\}$~~ (10) and (14) $\Rightarrow \{C, D\} \rightarrow \{A, C\}$. Now transitivity with (16).
- 22 $\{B, D\} \rightarrow \{A\}$ item $\{A, C\} \rightarrow \{B\}$
- 23 $\{A, C\} \rightarrow \{E\}$

So we get one minimal cover:

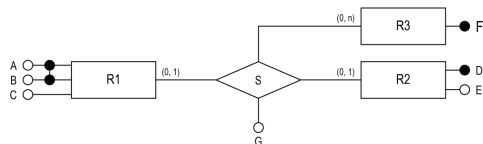
$\{\{C, D\} \rightarrow \{B\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B\}, \{A, C\} \rightarrow \{E\}\}$

Q 2.

Consider the following entity-relationship diagram. Find all functional dependencies that hold on R .



Q 2. Solution



- From $R1$ we have $\{A, B\} \rightarrow C$.
- From $R2$ we have $\{D\} \rightarrow E$.
- From $R3$ we do not have anything extra.
- From participation of $R1$ and $R2$ in S , $\{A, B\} \rightarrow \{D\}$ and also $\{D\} \rightarrow \{A, B\}$. We also have $\{A, B\} \rightarrow \{G\}$ and $\{D\} \rightarrow \{G\}$.
- From participation of $R1$, $R2$ and $R3$ in S , we have $\{A, B\} \rightarrow \{F\}$, and $\{D\} \rightarrow \{F\}$.

Codes

You can take a look at this repository that will help you create random FD problems for your practice, and also given a csv file, you can find functional dependencies:

https://github.com/pratik2358/fucntional_dep

Questions?

Drop a mail at: pratik.karmakar@u.nus.edu